Fiscal Year:	FY 2008	**Task Last Updated:**	FY 10/17/2008
PI Name:	Weichman, Peter B Ph.D.		
Project Title:	Theoretical investigations of equilibrium and nonequilibrium superfluid dynamics with application to future ground-based and microgravity experiments		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Low temperature condensed matter physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Burlington	**State:**	MA
Zip Code:	01803-5024	**Congressional District:**	6
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2002 Physical Sciences 02-OBPR-03
Start Date:	04/01/2004	**End Date:**	08/31/2008
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JPL
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Flight Program:			
Flight Assignment:	NOTE: No cost extension to 8/31/2008, per PI (10/07) NOTE: No cost extension to 9/30/2007, perhaps beyond, per PI (10/06)		
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):			
Grant/Contract No.:	GSN 4497		
Performance Goal No.:	1.24		
Performance Goal Text:	To explore the frontiers of physics and unlock the mysteries of the universe obscured by Earth's gravity		

Task Description:	Various aspects of the theory of equilibrium and nonequilibrium superfluidity will be considered with special emphasis on properties relevant to proposed and planned future microgravity experiments, and their ground-based precursors. Of special interest are steady equilibrium and nonequilibrium interfacial states, in which gravity (equilibrium) and heat current (nonequilibrium), or a combination of the two, supports coexistence of superfluid and normal phases with a well defined interface between them. Approximate mean field and (where feasible) more accurate renormalization group calculations will be performed (a) to extend present understanding of the steady state structure of these interfaces, and (b) to understand their near-steady state dynamics under the influence of various experimental probes, such as second sound. Also of interest are further investigations into the nature of the so-called self-organized critical (SOC) state, which supports a finite temperature gradient in the superfluid phase by generating a stream of vortex lines propagating across the direction of heat flow. I will investigate the fluctuations and spatial correlations in this state, its associated nonequilibrium specific heat, and the effects of a superfluid-SOC interface on measurements of the heat capacity at constant heat current.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This is a fundamental theoretical physics research project and impacts life on Earth only insofar as it pushes back the boundaries of knowledge and contributes to NASA's basic science mission.
Task Progress:	(1) In previous work, Dr. Weichman and his collaborators considered the large-scale, long-time dynamics of the heat-current-induced superfluid-normal interface. They showed that an interfacial second sound mode exists, in which an initial perturbation to the interface position will propagate at a well defined speed C(0)(S). This prediction has never been experimentally verified. Dr. Weichman has now proposed an "interfacial spectroscopy" experiment which should show an unambiguous signature of the interface mode. Thus, if a second sound pulse is propagated toward, and scattered by the interface, it is predicted to be almost entirely absorbed when the horizontal component of its phase speed matches C(0)(S). The observed reflected amplitude then shows a strong dip as a function of pulse frequency, with width proportional to interfacial dissipation constant D(0)(S). A paper describing these results entitled "Spectroscopy of a nonequilibrium superfluid-normal interface" has been published by the PI in Physical Review Letters, 96, 135301 (2006). (2) Dr. Weichman (with collaborators at University of Colorado, Boulder) has performed theoretical investigations of dilute Bose condensates with tunable pair interactions. As the interactions are tuned from repulsive to attractive, the atoms in the condensate form molecular pairs, which themselves may condense. The resulting phases and phase transitions have been described in the publication: L. Radzihovsky, J. Park, and P. B. Weichman, "Superfluid transitions in bosonic atom-molecule mixtures near a Feschbach resonance," Phys. Rev. Lett. 92, 160402 (2004). A much more detailed write-up was published in: L. Radzihovsky, P. B. Weichman, and J. Park, "Superfluidity and phase transitions in a resonant Bose gas," Annals of Physics 323, 2376 (2008). Basic research on magnetically trapped Bose condensed vapors, including Feschbach resonance phenomena, has been funded by the NASA Fundamental Physics Program since microgravity could lead to improved versions of these experiments due to the greater trap stability. (3) Although peripheral to this project, some related work should be mentioned on the onset of superfluidity in porous media---known as the "dirty boson problem." NASA has previously funded work on superfluidity in restricted geometries, and this work could eventually have relevance to that field. The work relates to the quantum critical dynamical properties of the phase transition, in which it is shown that a previously derived equality between the dynamical exponent z and the spatial dimensionality d can in fact fail. The mechanism for the failure, based on the presence of a broken particle-hole symmetry at the transition, may explain recently observed violations of the equality in two-dimensional quantum Monte Carlo simulations. The work is described in the publication: P. B. Weichman and R. Mukhopadhyay, "Critical dynamics of the dirty boson problem: revisiting the equality z=d," Phys. Rev. Lett. 98, 245701 (2007). A much more detailed write-up has been published in: P. B. Weichman and R. Mukhopadhyay, "Particle-hole symmetry and the dirty boson problem," Phys. Rev. B 77, 214516 (2008). This last paper was selected by the editors of PRB to be an Editors' Suggestion, one of a small number of PRB papers selected from each issue that the editors and referees find of particular interest, importance, or clarity. (4) Although not yet completed, Dr. Weichman has also begun a detailed study of the self-organized critical state, and related heat-current-induced phases, in the n --> infinity, or spherical limit, where n is the number of order parameter. This limit, first introduced about a decade ago by R. Haussmann, provides an exactly soluble limit in which various questions about nonequilibrium superfluidity can be explored. The advantage of this limit, over conventional mean field theory, for example, is that it automatically provides for dissipation in the superfluid phase. Although the nature of this dissipation may be rather different for n > 1 than for n=1 since there is no topological energy barrier to phase slips, which at n=1 require nucleation of vortex rings, the approximation provides a number of interesting physical insights. The aim of the present work is to extend Haussmann's analysis to explore various time-dependent phenomena. The work perfomed thus far is described in an incomplete manuscript by the PI entitled "Nonequilibrium Superfluid Critical Dynamics in the Spherical Limit."
Bibliography Type:	Description: (Last Updated: 01/12/2021)
Articles in Peer-reviewed Journals	Radzihovsky L, Weichman PB, Park JI. "Superfluidity and phase transitions in a resonant Bose gas." Annals of Physics 2008 Oct;323(10):2376-451. http://dx.doi.org/10.1016/j.aop.2008.05.008 , Oct-2008
Articles in Peer-reviewed Journals	Weichman PB, Mukhopadhyay R, "Particle-hole symmetry and the dirty boson problem," Physical Review B 2008 Jun;77(21):214516. http://dx.doi.org/10.1103/PhysRevB.77.214516 , Jun-2008
Articles in Peer-reviewed Journals	Weichman PB. "Dirty bosons: twenty years later." Modern Physics Letter B. 2008 Oct 30;22(27):2623-47. https://doi.org/10.1142/S0217984908017187 , Oct-2008
Articles in Peer-reviewed Journals	Weichman PB, Mukhopadhyay R. "Critical dynamics of the dirty boson problem: revisiting the equality z=d." Phys Rev Lett. 2007 Jun 15;98(24):245701. PMID: 17677974 , Jun-2007

Articles in Peer-reviewed Journals	Weichman PB. "Second-sound spectroscopy of a nonequilibrium superfluid-normal interface." Phys Rev Lett. 2006 Apr 7;96(13):135301. https://doi.org/10.1103/PhysRevLett.96.135301 ; PMID: 16711998 , Apr-2006
Significant Media Coverage	Weichman PB, Mukhopadhyay R. "'Particle-hole symmetry and the dirty boson problem,' was selected by the editors of Physical Review B to be an “Editors Suggestion,“ one of a small number of papers selected from each issue to be of particular interest, importance, or clarity." Physical Review B 2008 Jun;77(21):214516. , Jun-2008